Luddy Undergraduate Research: Fall 2024
Student-Mentor Research Agreement

Scholar name: **Lucas Gottdenker**
Scholar email: **lgottden@iu.edu**
Scholar ID: **2001183847**

Mentor (PI) name: **Mitja Hmeljak**
Mentor email: **mitja@iu.edu**

Co-Mentor (if any):
Co-Mentor email:

Credit hours: (default: 1)

> **Notes:** At IU Bloomington, a credit hour equates to 1 hour of instruction and 2 hours of student work per credit hour for approximately 15 weeks (BL-ACA-H13), constituting a total of 45 hours. In the context of mentored Undergraduate Research, we combine these for a total of 45 hours of student work per credit hour of research. Scaling this over 12 weeks (start of Week 3 to end of Dead Week), students and mentors should match their research expectations to 3.5-4 hours of work per week per credit hour, *including student research time, mentoring, and lab meetings.*
>
> Mentors and students are encouraged to choose a "bite-sized," tangible advance towards an ongoing research project. Moreover, to encourage responsible research conduct, we recommend research objectives that *do not rely upon a specific research outcome* (e.g., prove that the Earth is an icosahedron), but rather emphasize performing an investigation or following a well-defined plan (e.g., evaluate seismography data with new analysis techniques to determine earth's shape).

Overall title (or theme):
Music Visualization

Specific research objectives / work for Fall 2024:
First and foremost, I will need to familiarize myself with the Unity game engine, as well as the C# programming language, in order to effectively use this platform to build a music visualization demo. I will also need to figure out what type of visualization technique I would like to replicate and determine how said technique can be translated into code. Finally, I will need to build a working demo and figure out what meaningful information can be gleaned from its representation of music (i.e. articulate why it is helpful/interesting). In particular, I would like to explore isochord visualizations as tools for harmonic analysis. I feel that isochords could be immensely beneficial in music education settings; they would provide students with an entirely new framework for processing harmonic relationships in the context of Western music theory.

Metrics for success (How will we decide if the student was successful?):
I would consider my work successful if, by the end of the semester, I am able to produce some kind of working music visualization demo in Unity. Considering that I am currently unfamiliar with Unity and C# (and quite new to programming in general), I think it is important that I keep my expectations at a reasonable level. Ideally, I would like to create a visualization project involving isochords, either as a means of visualizing harmonic information from a MIDI file, or as a tool for visualizing harmony during live performance (via MIDI controller). I would also like to be able to showcase this project in an interactive setting and provide a general overview/explanation of isochord visualization as a tool for intuitive harmonic analysis.